# Neel Shah

## Python Backend Development · FastAPI · Testing

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## SUMMARY

I build Python services and the tests around them. My project, CourtVision, runs a **FastAPI** layer of **~99 endpoints across 12 routers** (REST, **WebSocket**, **SSE**), 9 long-running **daemons** under a **watchdog**, **PostgreSQL** persistence with a **SQLite** fallback, a ~7,400-test **pytest** suite, CI/CD on **GitHub Actions** and 5 **Docker** images. Earlier, at Fortrex Securities, I refined **backend payment APIs** and reporting pipelines for high-throughput billing.

## SELECTED PROJECT: COURTVISION

*Calibrated multi-sport forecasting engine, with the validation harness and paper-only execution controls as first-class components.*
*Evidence: github.com/neeljshah/court-vision/blob/master/docs/QUANT_BRIEF.md · github.com/neeljshah/court-vision/blob/master/EVIDENCE.md*

**Backend and services**

- **API and services: FastAPI** serving layer of **~99 endpoints across 12 routers** (REST, **WebSocket**, **SSE**) and server-rendered Jinja dashboards; 9 long-running **daemons** under a **watchdog** supervisor.
- **Persistence: PostgreSQL** schema (12 core tables) with a transparent **SQLite** fallback that auto-translates PG SQL, plus tracked, idempotent migrations.
- **Testing and delivery:** ~7,400-test **pytest** suite across ~580 files, CI/CD on 3 **GitHub Actions** workflows, and 5 **Docker** images; a 430-module codebase with research code kept apart from the load-bearing deployment graph.
- **Execution controls, paper only:** an order lifecycle whose live path is unreachable behind three independent refusals; a venue fee model that **fails closed**; a CLV ledger with an off-market-price filter, with its writer tested **race-safe across two OS processes**. The forward paper series began 2026-10-06 (0 of 8 required weeks).
- **MCP and honesty CI:** an **MCP server** (interface documented publicly) answers queries from AI agents with cited artifacts; the build fails when a retracted figure appears outside retraction framing.

**Data, models and validation**

- **Harness:** expanding-window walk-forward with a per-fold date assertion, truncation-invariance leak tests, acceptance only when a change beats baseline on two independent corpora, and a published minimum detectable effect for every market-relative row.
- **Modeling and data:** per-stat XGBoost with isotonic calibration; four de-vig methods from scratch incl. Shin (1992); a 151-file intelligence layer incl. a 291,625-pair player-vs-player matchup matrix; AI agents propose signals and the gate rejects them.
- **Evidence site:** one unified, CI-gated site at https://neeljshah.github.io/court-vision/, built by one person directing AI coding agents; performance-budget and accessibility (axe) gates are hard CI steps with zero serious or critical violations; a PAPER ONLY desk record with receipts.

## PROFESSIONAL EXPERIENCE

**Data Analyst Intern, Sunsolor** · *Tempe, AZ · 2025*
Built live Tableau / Power BI suites; cut manual reporting latency ~30%; automated daily market-trend pipelines in Python; data-quality audits across 5+ datasets; optimized SQL on 1M+-row datasets.

**Data Annotation Specialist, DataAnnotation.Tech** · *Remote · 2024–2025*
Labeled 10,000+ structured samples for production ML training & LLM fine-tuning; authored corner-case documentation cutting pipeline error variance ~20%; refined labeling schemas with data scientists.

**Business Intelligence Analyst, Fortrex Securities** · *Madison, WI · 2023–2024*
Built secure executive reporting interfaces (99.9% uptime); refined **backend payment APIs** & reporting pipelines for high-throughput billing; engineered real-time anomaly-detection alerting.

## EDUCATION & CERTIFICATIONS

**University of Iowa**: three years of data-science coursework. Coursework: math stats, probability theory, machine learning, financial analytics.

**Certifications:** Certified Data Scientist (USDSI) · ML Specialist & Data Viz (Kaggle) · CBIAP (IABAC).

## SKILLS

**Languages:** Python (expert), SQL, JavaScript/TypeScript (working)

**Backend/Infra:** FastAPI, WebSocket/SSE, PostgreSQL, SQLite, Docker, GitHub Actions, GCP, Tableau, Power BI, dashboards

**Methods:** walk-forward / leak-free validation, calibration (isotonic), de-vigging (Shin), fractional-Kelly, Monte-Carlo simulation, shadow-logging, multi-agent orchestration

**ML/CV:** PyTorch, OpenCV, YOLOv8/Ultralytics, ONNX, TensorRT, NumPy, Pandas, scikit-learn, XGBoost, LightGBM, statsmodels